

AWS Target Architecture - mirrors Azure labels and flows

Azure labels to AWS equivalents

- Global Traffic Manager -> Route 53 latency routing
- Azure Front Door + WAF -> CloudFront + AWS WAF
- Azure CDN endpoint + SAS -> CloudFront OAC + S3 signed URLs
- Azure App Service -> EKS or ECS services
- Azure AI Search -> OpenSearch Service
- Azure Key Vault + MSI -> Secrets Manager + KMS + IAM roles
- Azure VNet + Private DNS -> VPC + Route 53 private hosted zones
- Azure Blob private container -> S3 private buckets block public access

Users and access

- Boeing user/admin/contractor/anonymous
- PingFed WSSO and FedProxy -> OIDC and SAML
- Hivebrite JWT for R/W APIs; R/O APIs anonymous

Client apps and ingress

- Hivebrite Safety Experience iframe -> Timeline Web App
- Boeing.com iframe -> Timeline Web App
- FirstUp RSS feed -> Timeline API Service
- Route 53 latency -> CloudFront + WAF -> ALB per region

Region A us-west-2 VPC boundary: bdse-services-vpc

- Public subnets: public-a/public-b -> ALB + NAT Gateway
- Private app subnets: app-a/app-b -> Timeline Web App, Timeline API
- Private app subnets: app-a/app-b -> Search Web App
- Private data subnets: data-a/data-b -> Neptune writer + reader
- Private data subnets: data-a/data-b -> Aurora PostgreSQL writer
- Private data subnets: data-a/data-b -> OpenSearch domain
- Private data subnets: data-a/data-b -> ElastiCache Redis
- VPC endpoints: S3, Secrets Manager
- Private DNS: Route 53 private hosted zone for DB endpoints
- Security: SG + NACL; monitoring: CloudWatch + X-Ray
- SKUs initial: Neptune db.r6g.2xlarge writer and reader
- SKUs initial: Aurora PostgreSQL db.r6g.xlarge writer
- SKUs initial: OpenSearch r6g.large.search data x3, masters x3
- SKUs initial: ElastiCache Redis cache.r6g.large

Region B us-east-1 VPC boundary: bdse-services-vpc

- Public subnets: public-a/public-b -> ALB + NAT Gateway
- Private app subnets: app-a/app-b -> Timeline API, Search API
- Private data subnets: data-a/data-b -> Neptune replica or async
- Private data subnets: data-a/data-b -> Aurora PostgreSQL replica
- Private data subnets: data-a/data-b -> OpenSearch replica
- S3 cross-region replication for static and media
- Route 53 failover and CloudFront origin group

DevOps VPC boundary: bdse-developer-vpc

- devops-subnet -> EC2 Linux workstation for data loads to PostgreSQL
- VPC peering or Transit Gateway to bdse-services-vpc

User categorization Lambda -> Hivebrite Admin API